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Status and initial experiments of DRAGON-V facility for lithium-lead blanket technologies of hydrogen fusion reactors

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HIGHLIGHTS

- A lithium-lead facility with 40 kg/s mass flow and 2–5 T magnetic field was built.
- The composition and performance of the device are described and tested in detail.
- A novel cold trap system with three cooling zones has been designed and proposed.
- The safety analysis of loss of flow accident and loss of heat sink was investigated.

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ABSTRACT

A dual-coolant integrated experimental facility named DRAGON-V has been developed at the Institute of Nuclear Energy Safety Technology, Hefei Institute of Physical Science, Chinese Academy of Sciences, for the key technology research and performance evaluation of candidate liquid lithium-lead (PbLi) blanket of hydrogen fusion reactors. The loop is composed of a material test sub-loop and thermal-hydraulic test sub-loop, the design parameters are PbLi inventory 20 tons, PbLi temperature up to 550 °C, the maximum PbLi flow rate up to 40 kg/s. A novel cold trap system is designed to remove the suspended and crystalized impurities in PbLi fluid with three cooling zones and cross row arrangement of rod bundle filter elements. The paper describes the loop itself and its major components, initial loop testing, flow and measurement diagnostics and current experiments. The obtained test results of the loop and its components have demonstrated that the new facility is fully functioning and ready for experimental studies of material corrosion with/without a magnetic field, magnetohydrodynamic (MHD) effect, purification, heat and mass transfer phenomena in PbLi flows and can also be used in mock-up testing in conditions relevant to fusion applications.

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Introduction

Fusion energy is generally considered as an ideal sustainable energy system to replace fossil fuels in the future [1]. The hydrogen isotopes adopted as nuclear fuels collide and fuse together to produce helium and release enormous amount of energy in the process [2]. Many nuclear fusion scientific research devices have been built around the world, and the grandest hydrogen fueled fusion device, International Thermonuclear Experimental Reactor (ITER), will be expected to be completed in 2025. The Breeding blanket is a key component for efficient heat transfer and energy extraction in a fusion reactor [3], and the liquid lithium-lead (PbLi) blanket is an attractive breeding blanket concept design [4,5] because it can lead to tritium self-sufficiency without additional neutron multiplier, is immune to irradiation damage, has a high thermal conductivity and allows an in situ adjusting of the breeding rate by modifying on-line the concentration, etc. [6]. A particular blanket system called DCLL (dual-coolant lithium-lead) in the US [7], WCLL (water cooled lithium-lead) in Europe [5] and DFLL (dual-functional lithium-lead) blanket in China [8] have been proposed that high-temperature liquid PbLi flows slowly to remove the volumetric heat generated by neutrons and produce tritium, while a pressurized helium gas is used to remove the surface heat flux and cool the first wall.

Owing to the harsh environment of the PbLi blanket, such as high thermal load, great pressure gradient and high-energy neutron irradiation, lots of key issues such as material compatibility [9,10], thermal-hydraulics performance [11,12], tritium permeation [13,14] and safety analysis [15,16] etc., should be investigated and paid wide attention. Therefore, for the sake of investigating comprehensively the above issues, the out-of-pile test platforms are necessary and the experiments should be performed in PbLi experimental facilities before their ultimate application in reactors.

At present, lots of PbLi experimental facilities are being constructed and successfully operated worldwide. To understand the hydrodynamic properties of PbLi flows, some purely hydrodynamic loops without magnetic field have been built, such as PICOLO at KIT in Germany [17], ANAPURNA at CEA in France [18], LIFUS II at ENEA in Italy [19], and a high-temperature PbLi loop at Kyoto University in Japan [20]. Besides, a few magnetohydrodynamic (MHD) PbLi facilities were constructed to investigate the magnetic field effects: a Pb–17Li loop at IPUL in Latvia [21], the ELLI loop at the Korea Atomic Energy Research Institute [22], the MaPLE loop at UCLA in U.S [11] and the LIMITEF 5 facility at JSC “NIIEFA” in Russian with superconducting magnet (up to 5.5 T) [23]. In China, several PbLi loops, named DRAGON I-IV have been designed and fabricated in the Institute of Nuclear Energy Safety Technology (INEST), Hefei Institute of Physical Science, Chinese Academy of Sciences [24].

Using these facilities, many studies have been conducted until now, e.g., compatibility and corrosion behavior of candidate steels [25,26], chemical interaction between liquid PbLi and water [27], and compatibility of silicon carbide (SiC) with high-temperature alloys [28]. Although good knowledge of PbLi behavior has been achieved, many practical problems vital to liquid lithium-lead blanket remain open, including MHD and heat transfer tests of the TBM mockups, tritium permeation barrier (TPB), lithium-lead impurity control and purification, and design basis accident analysis (LOCA, LOFA, etc.) [29] and so on. Thus, further studies are required.

Actually, there are few existing forced-convection PbLi experimental facilities with high flow rates and strong magnetic field, such as 2.65 L/min without a magnetic field in the PICOLO loop, 50 L/min with a uniform magnetic field up to 1.8 T in MaPLE, and approximately 66.7 L/min with a magnitude of 3–5.5 T in LIMITEF5, etc. The obtaining experimental findings cannot be fully extracted and match the requirements of the engineering design of a fusion reactor blanket. To support the engineering design validation of the PbLi blanket and to perform key issues and integral experiment verification such as the compatibility of PbLi with Chinese RAFM steel CLAM, MHD effect test in complex ducts, and He/PbLi dual-coolant thermal-hydraulics test in the multi-physical environments, a dual-coolant thermal-hydraulic integrated experimental loop called DRAGON-V with a maximum flow rate of roughly 252 L/min (40 kg/s), 550 °C and 2–5 T magnetic field was developed [30] in INEST for ITER DFLL-TBM [31] and future China DEMO blanket [32]. In this article, it will mainly focus on introducing more details about the PbLi loop of the facility, including major components, measure and flow diagnostics, impurity control, performance testing, its operation and some of the results of current experiments.

Experimental setup

Description of the facility

PbLi loop

DRAGON-V facility consists of two parts: PbLi loop and Helium circuit, of which the main PbLi loop (see in Fig. 1) was constructed at INEST in 2017, and the helium circuit has completed the design. As a diagram shown in Fig. 2, the main PbLi loop is composed of two flow circuit, named material test sub-loop and thermal-hydraulic test sub-loop located on the left and right of the figure, respectively, and each of them can be in operation independently. The main components of this

Fig. 1 – Overall view of DRAGON-V PbLi loop.

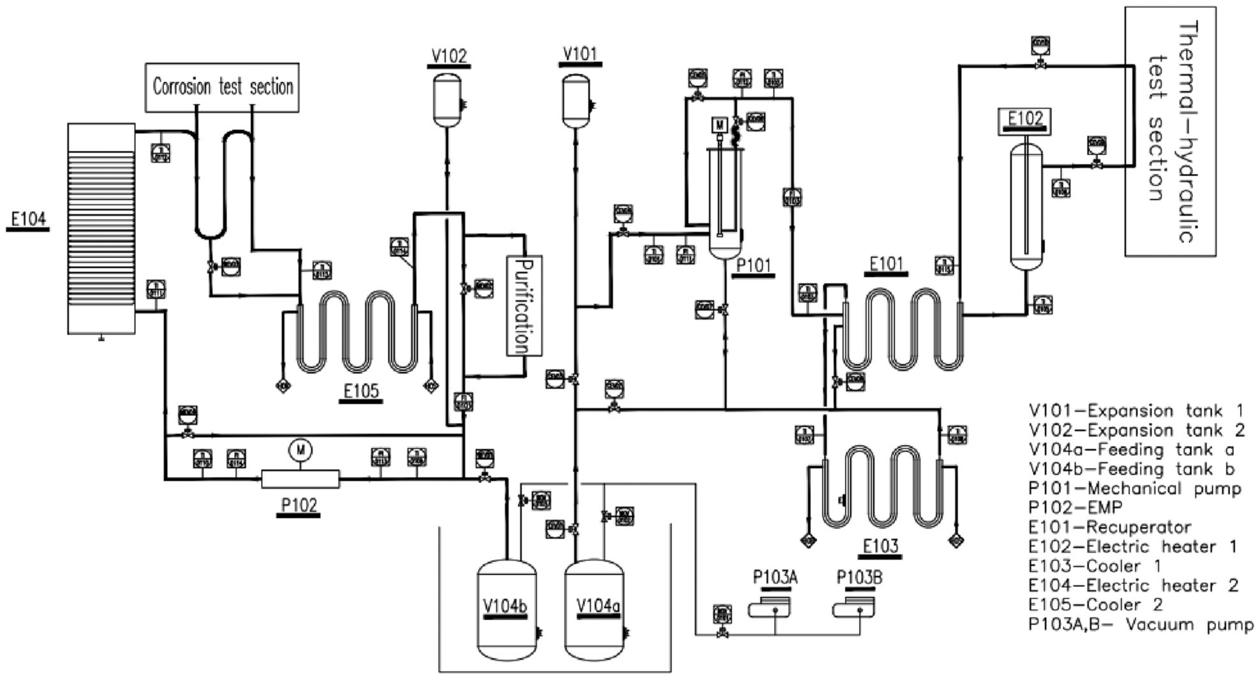

Fig. 2 – DRAGON-V P&ID.

loop include melting and feeding tanks, electromagnetic pump (EMP) and mechanical pump, two flowmeters, electromagnet, two electric heaters E102 and E104 and two coolers E103 and E105, etc. The main parameters are the following: overall dimensions of 8 (length) × 5.5 (width) × 2.6/3.5 (height) m, pipes material-stainless steel SUS 316L steel, main loop pipes diameter Ø45 × 4 mm and Ø89 × 6 mm, PbLi temperature range of 350–550 °C. The maximum inventory of PbLi in the loop is now about 20 tons.

The material sub-loop is mainly to investigate the material-related technologies, such as the corrosion behavior of candidate structural materials with and without the magnetic field, PbLi purification and hydrogen isotope extraction, etc. The molten PbLi in the feeding tank is pumped into the electrical heater E104 by EMP, and the liquid PbLi in E104 is heated and the outlet temperature can be up to 550 °C which is satisfied with the requested temperature of blanket in the test section. Then, the liquid lithium lead cooled by the heat exchanger E105 flows back to the pump. For PbLi impurities control and purification, a cold trap with wire meshes to filter and capture impurities is installed in the by-pass branch, the mass flow rate range of PbLi in this trap is 1.2–3.2 kg/s.

The thermal-hydraulic sub-loop is to study the thermal-hydraulic integrated experiments for the R&D of a liquid blanket, such as the MHD effect, 1/3 scaled TBM mockup test, thermal hydraulics and safety issues, etc. The main loop is an "eight-shape" loop, the fluid is pumped by mechanical pump into the electric heater E102. The heated fluid then flows into the cooler E103. Different from the material sub-loop, a recuperator placed between the cooler E103 and the heater E102 is used, one is to recover the enthalpy and on the other hand it can reduce the great thermal stress induced by the high temperature difference in the cooler E103. An auxiliary oil circulation loop is used and shared for the heat exchange of the cooler in thermal-hydraulic and material sub-loop.

Helium circuit

A diagram of the helium gas loop is shown in Fig. 3. The main circuit of the helium cooling system is designed as an "8" shape structure, which is divided into cold pipe section and hot pipe section. The regenerator is connected with the cold and hot pipe sections. The main components of the circuit include helium centrifugal compressor, regenerator, heater, cooler and filter, etc. The helium is driven into the regenerator by the compressor, then flows through the electric heater and the experimental section, and then returns to the regenerator. After that, the helium is heat exchanged through the helium-water cooler. After the helium is cooled, it returns to the inlet of the compressor through the filter, and finally realizes the circulation flow of the helium. The main design parameters of the helium gas loop are listed in Table 1. The filter is composed of a molecular sieve bed and an activated carbon bed, the molecular sieve bed is used to adsorb water vapor and carbon dioxide in helium, and the activated carbon bed is used to adsorb nitrogen, carbon monoxide and other impurity gases.

Pump

To pump PbLi, a Linear Induction Electromagnetic Pump (EMP) with power supply parameters 380 V, 150 A, 50 Hz manufactured by IPUL, Latvia, is used in material sub-loop and a mechanical pump with a maximum shaft power of 11.2 kW is installed in another sub-loop. The diameter and length of the EMP which is located on a height of 2600 mm platform shown in Fig. 4 are 240 mm and 2070 mm, while the length of the

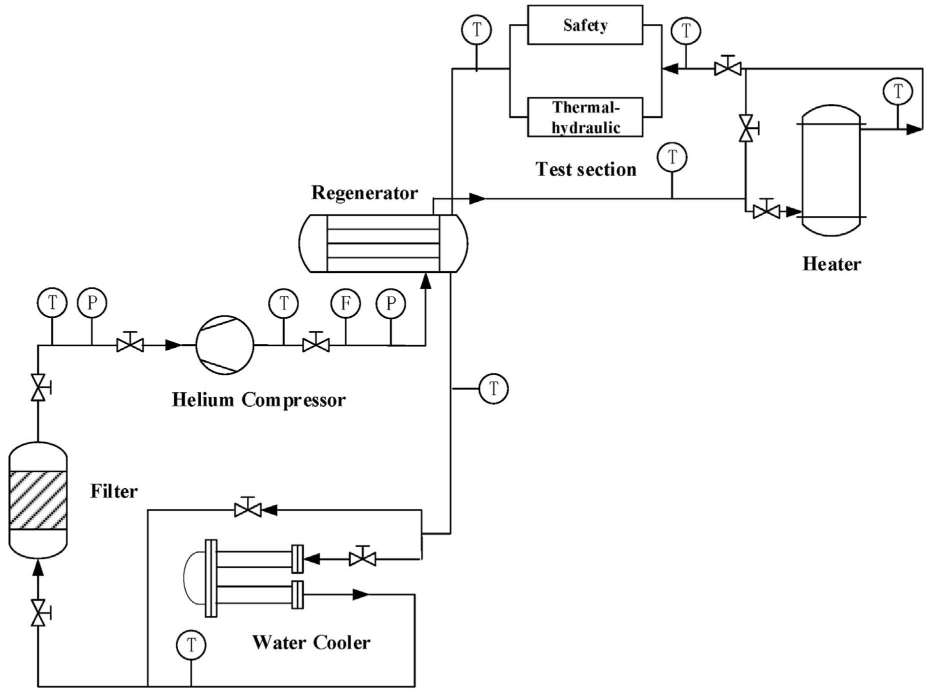

Fig. 3 – Schematic diagram of helium gas loop.

mechanical pump positioned on a 3500 mm platform is about 3000 mm.

The operating temperature of the pump in the loop should be controlled at 350–500 °C, so the EMP temperature measuring is necessary to protect insulation damage of the windings from unacceptably high temperature. Measuring of temperature distribution in EMP windings and on the input and output of the channel as well along its length. Eight K-type thermocouples are employed, of which three thermocouples are arranged in phase windings, two thermocouples mounted on the inlet and outlet surface of the channel, two thermocouples are installed on the water inlet and outlet pipes of the pump. Additionally, two K-type thermocouples inserted into the flow are used to detect the temperature of the PbLi flow in the pipeline connecting the inlet and outlet of the pump.

To the knowledge of the pressure difference between the inlet and outlet of the pump during the operation of the circuit, a Rosemount pressure transducer, which the operation temperature can be up to 410 °C, is installed at the inlet and outlet of the pump.

Electric heater

The DRAGON-V PbLi material sub-Loop electric heater E104 is a coil tube structure and the PbLi flows in the spiral tube and radiates heating by heating rods. The power can be adjusted between 0 and 100 kW, the coil tube material is T91, pipe diameter is Ø 45 × 4 mm, and the length is approximately 40 m. A total of 68 (U-tube configuration) electric heaters are arranged inside the heaters (2 splitting) with the dimension Ø 12 × 1.5 mm, and the surface load of each heater is 1.6 w/cm^2. DRAGON-V PbLi thermal-hydraulic sub-loop electric heater E102 is a cylindrical structure and the power can be adjusted between 0 and 70 kW. A bundle of 96 (U-tube configuration) electric heating rods are arranged in the container, of which 6 are alternate heating rods. The liquid PbLi fluid is in direct contact with the heating rods. The heating rod is Ø 12 × 1.5 mm and the single surface load is 1.58 w/cm^2. Two K-type thermocouples, which are bundled with electric heating tubes, are mounted inside each electric heater. It can be used to control the temperature of the electric heating tube for an over temperature protection and preheating stage.

Table 1 – Main parameters of the helium gas loop.

Pressure/MPa	10.5
Flow rate/kg/s	1.08
Temperature/°C	550
Power of heater/MW	0.55
Temperature of compressor/°C	150

Fig. 4 – EMP system in the PbLi loop of INEST.

The electric heater is divided into two phases during the operation of the PbLi loop, one is preheating, and the other is running. In the preheating phase, an initial target temperature needs to be set firstly and then the temperature from an internal K-type thermocouple is fed back to the heater. The final preheating temperature can be achieved in a coil tube or cylinder. At the running stage, as the coil or the body is filled with PbLi, the power supply is controlled and adjusted by setting the desired PbLi temperature provided at the K-type temperature sensor at the exit of the heaters, and the final experimental requirements of different PbLi temperatures are achieved.

Heat exchanger system

The heated PbLi should be immediately cooled down to avoid overheating downstream of the test section, the upper limit temperature is 550 °C. In this work, a concentric recuperator, E101 and two coolers, E103 and E105, were developed for the heat transfer of whole loop. A W-type structure is employed and the detail parameters of the recuperator have been introduced in Ref. 35. The two coolers have the two-layer W-type structure shown in Fig. 5, in which the heat exchange power of cooler E103 is 120 kW and that of cooler E105 is 70 kW. The material is S31668. The diameter of the tube side is consistent with the main pipes of each sub-loop and the diameters of the shell side in E105 and E103 are Ø 76 × 5 mm and Ø 133 × 6 mm, respectively.

Thermal oil is used as the coolant in the shell side of the coolers, which can be operated at a temperature of 300 °C avoiding the liquid PbLi solidification as mentioned above. By adjusting the flowrate and temperature of the oil, the outlet temperature of the liquid metal in the cooler can be controlled at the target value and the accuracy is ±5 °C, and the outlet temperatures of the two sets of cooler can operate and control independently but also run at the same time. In this work, the oil operates at 200 °C–300 °C and supplemental heating is required as the temperature is lower than 200 °C. A regulator module is used to control the heating process and the risk of heating wire is avoided due to full load heating by PID closed-loop adjustment. The heated oil is cooled by air-forced convection and radiation heat dissipation, the total heat transfer area is 26 m^2 and the heat transfer pipe is a fin tube with the dimensions of Ø 25 × 2 × 1000 mm.

Fig. 5 – Photograph of the cooler.

Corrosion test sections

For high temperature corrosion test of structural materials, a mock-up test section is design/manufacturing and installed at the exit of electric heater E104. The original mock-up consisting of several rectangular ducts with the material 316 Ti is a horizontal S-type structure which is facilitated by online sampling. The dimension is 30 × 30 mm inner diameter, 5 mm wall thickness. Experimental samples are cylindrical, and the front and rear ends of each sample are respectively processed with external and internal threads, and all the samples can be connected with each other one by one. The experimental samples are placed in two straight ducts for the comparative experiments, and one duct is placed for the tests of corrosion phenomena in a magnetic field environment. The intensity of the magnetic field created by a water-cooled electromagnet is 2 T, and the uniform space of the magnetic field is 800 mm (axial) × 200 mm (horizontal) × 80 mm (vertical). To perform the velocity measurement of lithium lead flow under the condition of magnetic field corrosion, the test section as shown in Fig. 6 has been modified with the dimensions of 15 × 15 mm recently and two high-temperature ultrasonic Doppler velocimetry (UDV) transducers developed in Switzerland (10 mm diameter and 300 mm length) are employed, but the experiment has not been carried out yet. In the next state, the magnetic field will be upgraded to 5 T.

Purification system

During high-temperature operation of heavy metal coolant reactors, the coolant will corrode and erode the structural materials. Number of corrosive impurities including suspended solid and dissolved impurities, appear in the coolant. On the one hand, the presence of solid impurities may scour the pipe wall and accelerate the corrosion of the material. On the other hand, the dissolved impurities are easily precipitated and crystallized on the immovable surface in a lower temperature section, which will increase the hydraulic resistance and even block the pipeline. In our previous experiments, the possible impurities in the cold trap are Fe, Cr, Ni

Fig. 6 – High temperature corrosion test section.

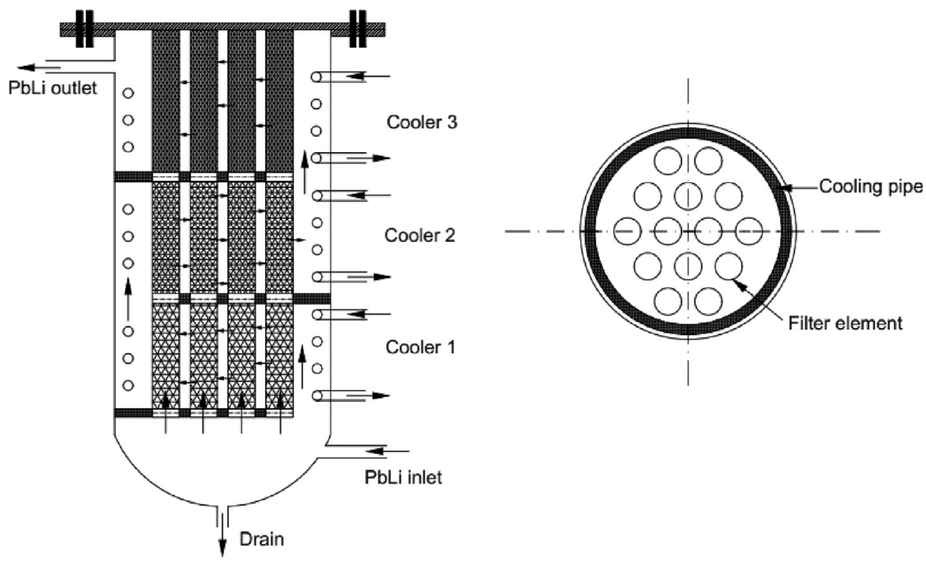

Fig. 7 – A modified Cold trap configuration.

and Mn, etc., and a small amount of PbO and Li_2O in the filter. The metallic elemental impurities of Fe, Cr, Ni and Mn, etc. mainly come from the corrosion of 316L stainless steel in liquid PbLi, the metal oxide impurities mainly come from the surface oxide of the bulk PbLi ingots and some amounts of free oxygen in the ingots. Therefore, the purification and deposit removal of liquid metal coolant are crucial for the normal and safety operation of the circuit.

A modified cold trap system was designed to remove suspended and crystalized impurities on the immovable surfaces by PbLi cooling. It is a crucial segment for a long-term safety and stably operation of high-temperature liquid lithium-lead loop. As shown in Fig. 7, three cooling zones using oil-cooling loop adopted to obtain a uniform temperature distribution in each horizontal cross-section, which can eliminate appreciable temperature gradients at the inlet and make an efficient operation of the trap. If the whole of the body of the trap is cooled at an equal rate over the entire height, rapid crystallization of impurities in the solution will lead to choking of the inlet of the trap. The temperature distributions of PbLi in cooler 1, cooler 2 and cooler 3 are 370 °C, 320 °C and 270 °C, respectively, with the mass flow rate range 1.2–3.2 kg/s. The dimension of the trap is Φ800 mm × 1500 mm and the retention time is in the range of 5–10 min. The filter structure material and screen material was 316L stainless steel.

In this cold trap design, the lower temperature at the outlet is defined as 270 °C, which is mainly considered from two aspects: firstly, the melting point of the PbLi is ~235 °C, to keep the bulk of PbLi in a molten state, the lower temperature limit of PbLi inside the cold trap is designated to be 270 °C, which can keep an acceptable safety margin from the solidification temperature. If the temperature of PbLi is reduced below 270 °C there is a possibility of plugging of PbLi in the cold trap due to the lower flowrate in a cold trap, which may further cause reduction of flow of PbLi through the cold trap. Secondly, the saturation concentration of metal impurities significantly decreases with the decline in temperature. For example, the saturation solubility of Ni and Mn at 370 °C are 2023 wppm and 145 wppm respectively [33]. When the temperature reduces to 270 °C, the saturation solubility of Ni and Mn reduces to 1059 wppm and 21 wppm, respectively. Most of the remaining quantity of Ni (964 wppm) and Mn (124 wppm) may have been precipitated inside the cold trap.

Rod bundle filter elements are integrated into the top cover and the screen filter element are employed and the top is connected to the flange cover by thread. When it needs be replaced, the top cover and the filter element are replaced as a whole, which is convenient for maintenance and replacement. The filter element is a hollow structure (see Fig. 8), and the fluid can flow upward from the inside, which not only enhances the filtering capacity but also reduces the flow resistance. Three different specifications of steel wire mesh are arranged in the three cooling zones. A coarse wire mesh with wire diameter 0.315 mm is arranged at the bottom layer to filter large impurities, including those suspended and precipitated on the surface, and a fine wire mesh with wire diameter 0.081 mm is arranged at the upper layer to deposit and filter precipitated crystalline impurities. The middle layer is between the two dimensions and the wire diameter is 0.193 mm.

A cross row layout of rod bundle filter elements was adopted in the cold trap, and it can improve the efficiency of

Fig. 8 – Filter element in cold trap.

impurity crystallization. Turbulent mixing will be formed when the liquid flows around the filter assembly, which will enhance the heat and mass transfer process of PbLi fluid. A supersaturated solution of impurities is generated due to PbLi cooling, which will lead to the crystallization of the impurities both on wire screen surfaces and in suspension. At present, the pressure drop of a single filter module was tested in a water circuit, and then according to the characteristic that the resistance coefficients of the same filter element in different environments is the same, the hydraulic pressure drop in liquid PbLi environment is obtained by conversion and the result is shown in Fig. 9. The performance tests of cold trap, including pressure drop, the rate of purification and the capacity of the trap will be carried out in subsequent experiments.

Flow measurements and oxygen control

Flow rate and pressure measurements

Flow rate measurement in molten PbLi is critical and exceptionally challenging because of the aggressive nature of PbLi and its high temperature. In this work, Venturi flowmeter is chosen and employed in the facility for flow rate measurement. According to the previous experimental experience in KYLIN-II facility, a significant signal drift and attenuation occurred in the process of long-term use of Venturi flowmeter with impulse pipe upward that is because there is some gas in the impulse pipe, which prevents the liquid metals directly contacting with the sensor diaphragm. To obtain an accurate differential pressure signal, the installation mode of the Venturi flowmeter is upgraded, the impulse pipes at both ends of the Venturi flowmeter are installed downward, which is convenient to effectively evacuate gases as the fluid enters the impulse pipe. Besides, the flowmeters are calibrated at the institute of Metrology.

In the facility, a commercial 3051S series pressure transducer manufactured by Rosemount, US, was used to monitor the pressure change at the inlet and outlet of the pumps, the operation temperature is up to 410 °C and the pressure transmitters can measure pressure up to 2 MPa.

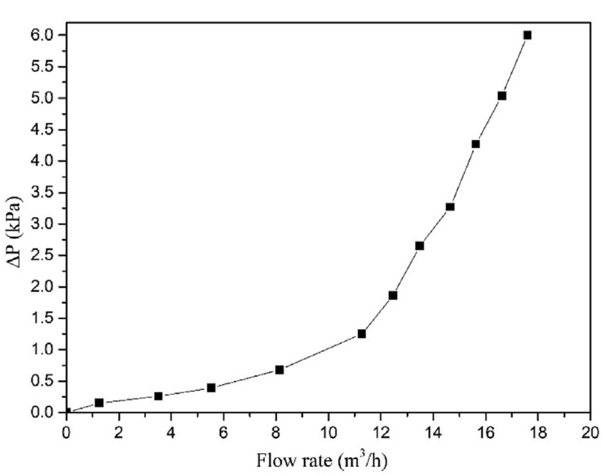

Fig. 9 – Flow rate vs pressure drop curve.

Temperature measurement

Temperature is a basic and probably the easiest physical quantity to measure. Thermocouples are widely available and can be commonly found. In the DRAGON-V facility, K-type thermocouples protected by a stainless steel sheath (SS 316L) are used, which can directly measure the temperature in the flowing PbLi between 250 °C and 550 °C. Most of them are installed directly on the outer surface of the pipes of the loop, while some thermocouples are inserted directly into the flow to detect the temperature of liquid PbLi in some designated positions, such as the inlet and outlet of heaters, coolers, pumps, etc. These thermocouples have been calibrated at the institute of Metrology and the uncertainty of the thermocouple is ±1 °C. On another hand, a handheld thermometer (Omega, HH11B) with 0.1% accuracy is applied as well to detect and judge whether the temperature measured by these thermocouples is accurate.

Oxygen control

As is known, oxygen content is one of the key factors affecting the corrosion and impurity of liquid PbLi. To monitor the oxygen content in the loop in real time, some independently developed oxygen sensors with a monitoring capacity limit of 10^{-9} wt% are mounted on the experimental test section in the loop, such as on the top of the glove box, purification experimental section and so on.

Additionally, to reduce and control the oxygen content in the entire circuit as much as possible, after loading the ingots into the melting tank, the air in the loop is evacuated using a vacuum pump. When the temperature of the melting tank and the whole loop increase to achieve a temperature of about 200 °C which the PbLi material is still solid in the melting tank, the vacuum pump is kept running and heating continues for 10–12 h until steady vacuum characteristics are achieved. When melting starts and the temperature is about 350 °C, vacuum pumping is continued. That is because some gas bubbles occur at the surface of the melt when PbLi is molten, which indicates that the gas bubbles (presumably of oxygen) are formed in the bulk liquid. Simultaneously, the entire loop is filled with argon at the pressure of about 0.1 MPa. When melting is finished, the liquid metal is pumped through the loop. Here, the vacuum pumping is alternated cycle with argon pumping to remove as much oxygen as possible. Finally, the whole loop operates under argon protection.

Results and discussion

To demonstrate the ability of DRAGON-V, some tests including the heating performance of the electric heater, pump and flowrate measurement, etc. have been done in this work, and some thermal-hydraulic experiments have been carried out as well. Due to the influence of factors such as work arrangement and project plan change, the tests on thermal hydraulic sub-circuit have not been carried out temporarily at this stage. Therefore, this study mainly focuses on the analysis and discussion of the test and experimental results of the material sub-circuit. The relevant conditions of thermal hydraulic sub-loop have been met and relevant series

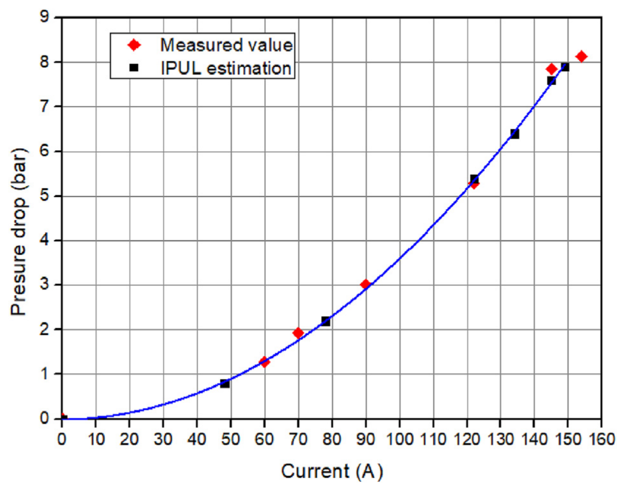

Fig. 10 – Testing results of pressure and current in DRAGON-V.

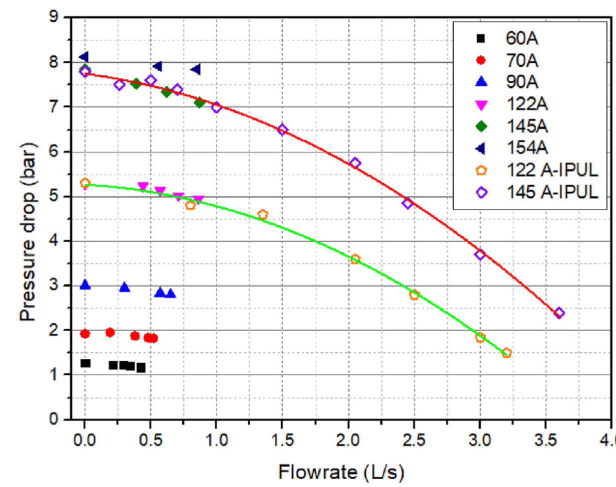

Fig. 11 – P–Q curve of EMP tested in DRAGON-V.

of tests and experiments have been put on the agenda and will be carried out in the near future.

Performance of the pump

To obtain the actual pump curve before starting experiments, it was necessary to conduct the performance test and the first run of PbLi flow in the loop was performed. At that time, the pump was continuously run for 2 days at a fixed pump voltage and current at a temperature around 300 °C to assure good wettability of the internal pump pipe and then another run continued for 4 days. In the whole process, it did not reveal any changes in the measured flow rate during the tests, suggesting good wettability conditions. A special series of runs was then performed to build the characteristic curve of the EM pump with different currents, which was varied from zero to the maximum one, such that the hydraulic resistance to the flow was significantly increased due to MHD effects. Fig. 10 shows the obtained pump curve (the developed pressure head as a function of the current) versus the manufacturer's estimation. Good accuracy was verified. From the pump performance tests, it has been confirmed that the maximum pressure in the output of EMP is increased to about 8.16 bar when current is up to 154 A.

The tested curve of the pressure head as a function of the flow rate is developed and shown in Fig. 11. It can be easily found that the test results agree with the pump performance curve obtained in IPUL. The maximum flow rate of 0.8 L/s and an output pressure of 7.9 bars can be achieved at the same time when the current increases to 154 A. It can be concluded that the facility has sufficient pumping capacity to perform various experiments related to the DFLL blanket concept in China.

Performance of the electric heater

In the DRAGON-V PbLi loop, preliminary heating test was carried out to check the performance of electric heater E104 in this paper. To achieve a full of power, a larger temperature difference is needed between the inlet and outlet of the electric heater. Here, setting the outlet temperature of the heater to 715 °C, which is more higher than the inlet temperature 308 °C, the heater is in a full of power operating, then recording the electric heater import and exit temperature values, the heated power is estimated from the differential temperature, flow rate, and the specific heat of PbLi [34]. In the whole test, the mass flow rate remains at 5.8 kg/s.

Fig. 12 shows the test power of electrical heater E104. In this operation, the highest outlet temperature of 406 °C was achieved and the inlet temperature was unchanged. The maximum temperature difference of lithium lead in the inlet and outlet of the heater was 96 °C and the corresponding heating power of the heater was about 105 kW that was higher than the design value of 100 kW. Note that the measured temperature is the actual temperature of the liquid PbLi because the two thermocouples are not attached to the outer surface of the pipe, but inserted into the flow. Besides, the result also indicated that the average heating rate was 5 °C/min.

Additionally, to meet the requirements of high temperature corrosion experiment, the high temperature performance

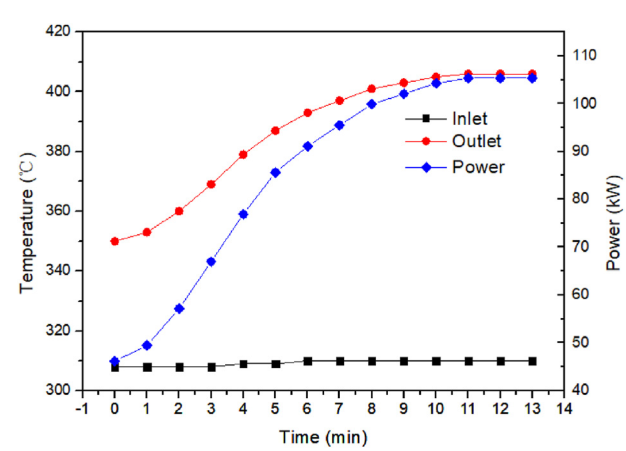

Fig. 12 – Heating power of electrical heater E104.

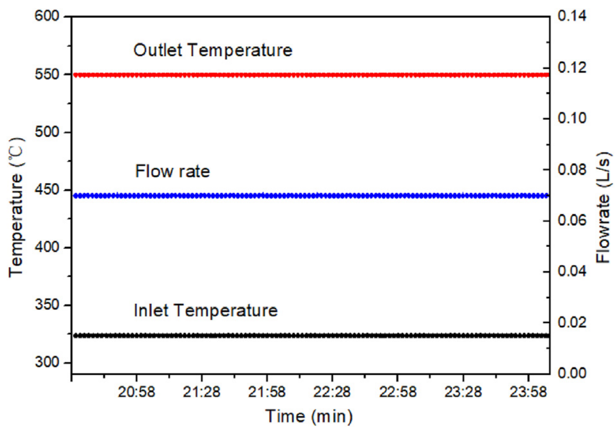

Fig. 13 – Temperature of PbLi in the inlet and outlet of heater.

of the loop needs to be tested. As shown in Fig. 13, the highest temperature of PbLi was achieved at 550 °C in the outlet of the heater with the nominal flow rate of 0.07 L/s in the loop and near 3 h stably operation keeping the temperature of 550 °C was carried out.

Performance of the heat exchanger

As mentioned above, the heat exchanger system consists of two coolers and a concentric recuperator in DRAGON-V. In previous studies, the thermal hydraulic and mechanical characteristics of concentric recuperator heat exchanger were analyzed by CFD code [35,36]. In this work, the experimental test of the heat transfer performance of cooler E105 is carried out under the condition of a maximum PbLi flow rate, 0.8 L/s. In this facility, when the entire loop operates stably and the loop heat loss reaches equilibrium, the lithium lead temperature at the inlet and outlet of the heater is approximately equal to the inlet and outlet temperature of the cooler. Therefore, it can be found that when the flow rate is maximum, the temperature of PbLi at the inlet and outlet of the heater shown in Fig. 14 is 272 °C and 334 °C, respectively.

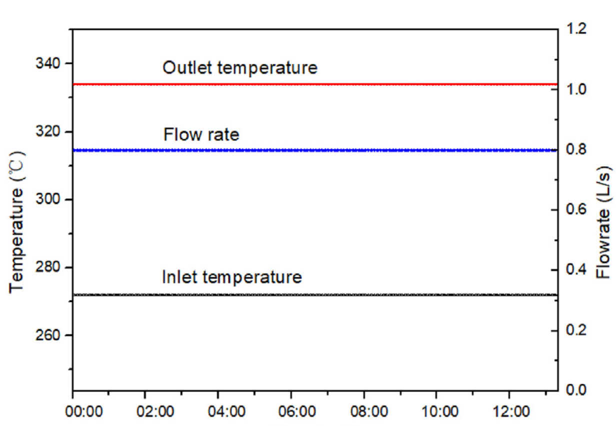

Fig. 14 – Temperature curve of heater under maximum flowrate.

The total heat transfer capacity of the cooler is nearly 90 kW that is higher than the design value of 70 kW and suggests that the cooler has high performance in heat exchange.

Preliminary thermal-hydraulic experiments

In a liquid PbLi blanket, transient safety events analysis including loss of coolant accident (LOCA), loss of flow accident (LOFA) and loss of heat sink, etc. is one a key and vital technologies that should be investigated widely. At present, most researchers focus on the investigation of in-box LOCA, including experimental and simulation analyzes [37–39]. In our previous work, many efforts were also made to investigate blanket transient behavior induced by the in-box LOCA [40–42], and some code was developed for accurately evaluating the safety of the blanket. However, the study of LOFA and loss of heat sink in a liquid PbLi blanket is rarely and the simulation code has yet to be developed and verified. Hence, some simulation experiments have been carried out in the facility to investigate the transient thermal-hydraulic behaviors induced by LOFA and loss of heat sink and the obtained experimental data can assist the development and verification of analytical code in the future.

Loss of flow accident

In this experiment, the electric heater E104 is taken as the simulation research object. Before the experiment, the circuit was in normal operation with a flowrate of 0.52 L/s, the outlet temperature of PbLi in the electric heater was 350 °C. Fig. 15 (a) shows that the flow rate of PbLi drops sharply from 0.52 L/s to 0.1 L/s in 5s when the pump driving force reduced to 50%, while the flow rate only decreases to 0.4 L/s in 5s as the driving force of the pump drops to 75%. The outlet temperature of PbLi in the electric heater is shown in Fig. 15 (b) raised rapidly from 350 °C to 400 °C in 16 min and the inlet temperature decreased from 309 °C to 280 °C under the condition of 50% pump driver, the corresponding average heating rate was 3.4 °C/min. When the driving force was 75%, the outlet temperature increased slowly from 350 °C to 365 °C in 8 min and the average heating rate is 1.25 °C/min, then the temperature keeps in equilibrium, indicating that the circuit has reached a dynamic thermal balance. Under the same experimental conditions, the more the flow decreases, the more significant the temperature increases. If the pump driving force is lower than 50%, the heating rate will become faster and the temperature will rise higher in a short time.

Comparison of the loss of flow accident happened in DFLL-TBM, the mass flow rate of PbLi is 4.33 kg/s (~0.45 L/s) in the design of DFLL-TBM [31], which is lower than the experiment flow rate of 0.52 L/s, but the nuclear heat deposited (~0.178 MW) carried away by the flowing PbLi in DFLL-TBM is higher than the total heat power of the electric heater E104 (0.105 MW). Therefore, if the flowrate in DFLL-TBM declined to 1/5 of the original flowrate corresponding to the condition of 50% pump driving force in the current simulation experiment, the temperature difference between the inlet and outlet of PbLi in the DFLL-TBM blanket will be about 5 times of the original design temperature difference (480/700 °C), which will lead to the temperature of FW increased significantly and maybe threaten the integrity of TBM. The current simulation

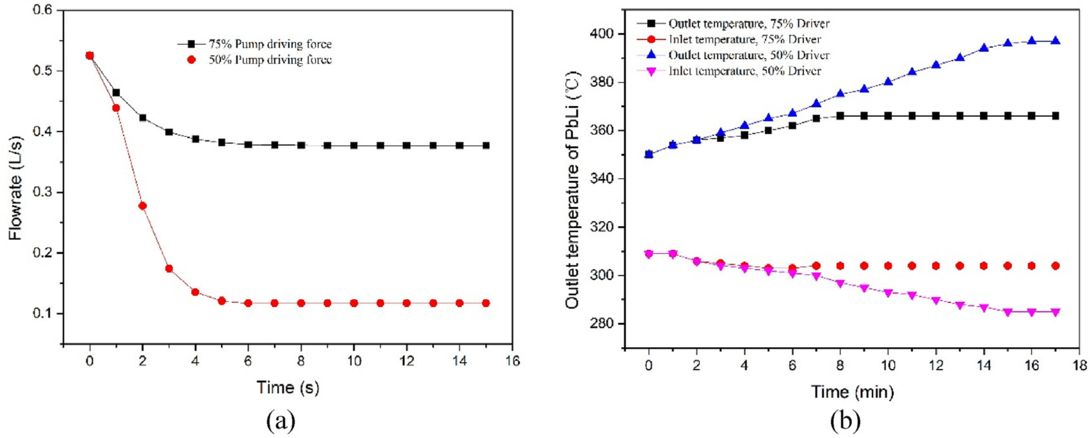

Fig. 15 – Flow rate (a) and outlet temperature of PbLi (b) curve vs time under different pump drivers.

experiment is a preliminary pre-test and more systematic and detailed experimental research will be carried out on the safety analysis of loss of flow accident in the next step.

Loss of heat sink

The safety experimental analysis for the loss of heat sink in the loop is carried out by cutting off the operation of the secondary oil pump and then monitoring and recording the temperature history of PbLi on the primary side and oil on the secondary side of the cooler. The initial experimental conditions were the same as above. Besides, the minimum outlet temperature of PbLi in the cooler is maintained at 270 °C to keep an acceptable safety margin from the solidification temperature (~235 °C) because the regulation process of the oil-cooled heat exchanger needs a certain response time. If the outlet temperature of the cooler is lower than 270 °C, the liquid PbLi may be excessively cooled and solidified when performing this transient condition experiment. The temperature evolution is shown in Fig. 16. When the outlet temperature of PbLi in the cooler increase from 294 °C to 336 °C and the temperature of cooler system reached a heat balance

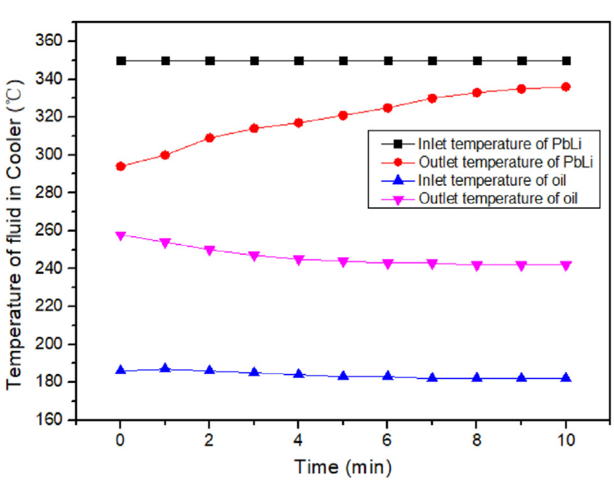

Fig. 16 – Temperature evolution vs time on both sides of the cooler.

within 10 min, the final temperature difference of inlet and outlet is maintained at 14 °C. The outlet temperature of the oil decreases slightly from 258 °C to 242 °C and the temperature difference keeps equilibrium after 5 min. The whole heat sink system reached equilibrium in 10 min. This illustrates that the lithium lead circuit has good regulation ability in a short time, even if a loss of heat sink accident occurs in the PbLi auxiliary system.

Conclusion

A dual-coolant integrated experimental facility named DRAGON-V for R&D activities of key issues of ITER DFLL-TBM and China DEMO was constructed at INEST. The loop is fully functioning and ready for experiments and mock-up testing. In this paper, the overall layout, key components and the measurements of the loop are introduced in detail and the performance of the material sub-loop and components has been tested.

1) A dual-coolant thermal-hydraulic integrated lithium-lead loop with PbLi inventory ~20 t, a maximum flow rate of 40 kg/s, 550 °C and 2–5 T magnetic field was developed to test the key R&D issues and validate the feasibility of PbLi blanket designs and the performance of the material sub-circuit, electric heater, heat exchanger and pump, etc. have been tested in this paper.
2) A modified cold trap with three cooling zone design is proposed, and some rod bundle filter elements with an integrated top cover and screen is adopted with cross row arrangement inside the trap. The temperature distribution in the inlet and outlet of the trap is 270 °C and 370 °C respectively.
3) The transient safety analysis of LOFA and loss of heat sink was pre-test preliminarily in this work. The results indicated that the temperature will rise higher in a short time when the LOFA and it maybe threaten the integrity of TBM.

Further experiments using this equipment will be carried out including magnetic field corrosion of structure materials, PbLi purification, MHD tests with higher magnetic field and temperature, DFLL-TBM mockup tests and transient safety analysis under conditions similar to those of a fusion power plant. We also plan major upgrades of the facility by coupling a helium gas loop to study the performance of heat transfer between PbLi and helium gas in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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